

TrafficBoard: Digital Spatio-Temporal Pinboard for Human Trafficking Detection

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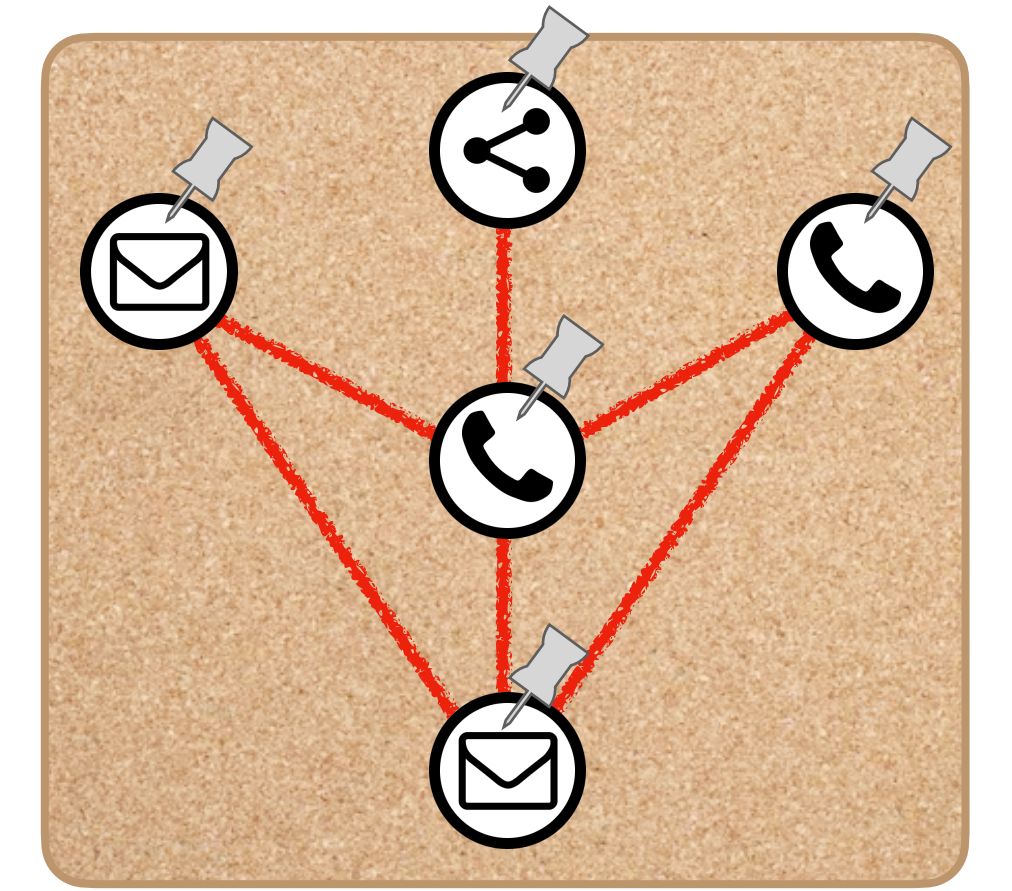
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ANALYTICS

Human trafficking for forced sexual exploitation (HT) affects an estimated 6.3 million people world-wide. Practitioners can detect HT in online escort websites by analyzing connections between metadata (emails, phone numbers, social media accounts, URLs) throughout time of related ads. They use *pinboards* (node-link diagrams), which get cluttered when parts of the graph are densely connected. **TrafficBoard** combines *graph compression techniques* and *interactivity* to enable practitioners to investigate connections between suspicious metadata over time, as well as their geographic spread.

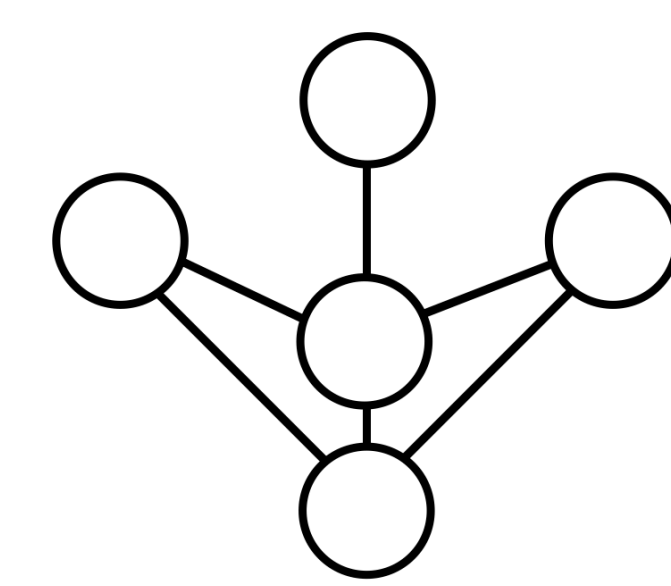
✓ *Summarizing the display graph*

✓ *Fast spatio-temporal exploration*

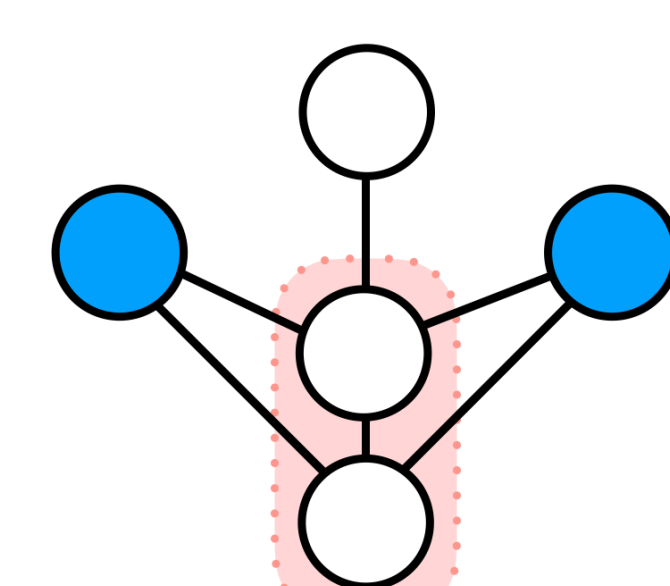
Example of a pinboard used by detectives:



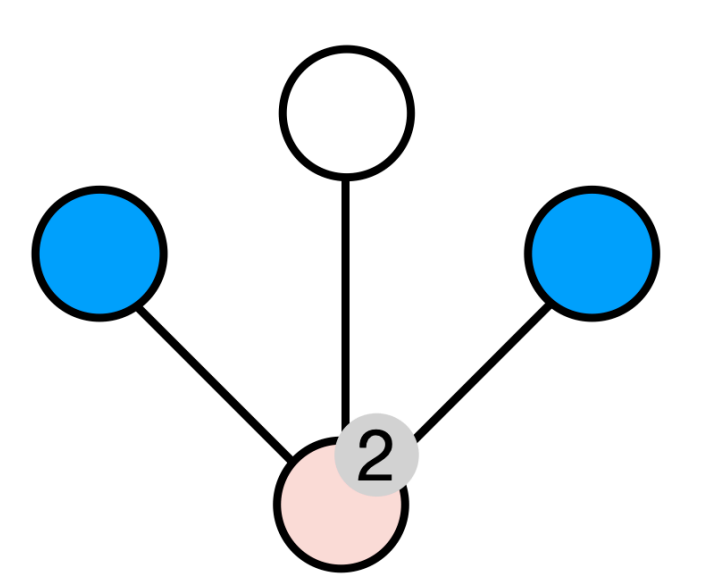
1 Graph compression. (a) First, we construct our graph by linking two metadata nodes if an ad uses both of them simultaneously. (b) We then pick the top 6 nodes (due to space constraints) with highest betweenness centrality and call them the *central nodes*. (c) Finally, we compress any non-central nodes with the same neighbor set into *supernodes*, resulting in our final display graph.



(a) Construct metadata graph

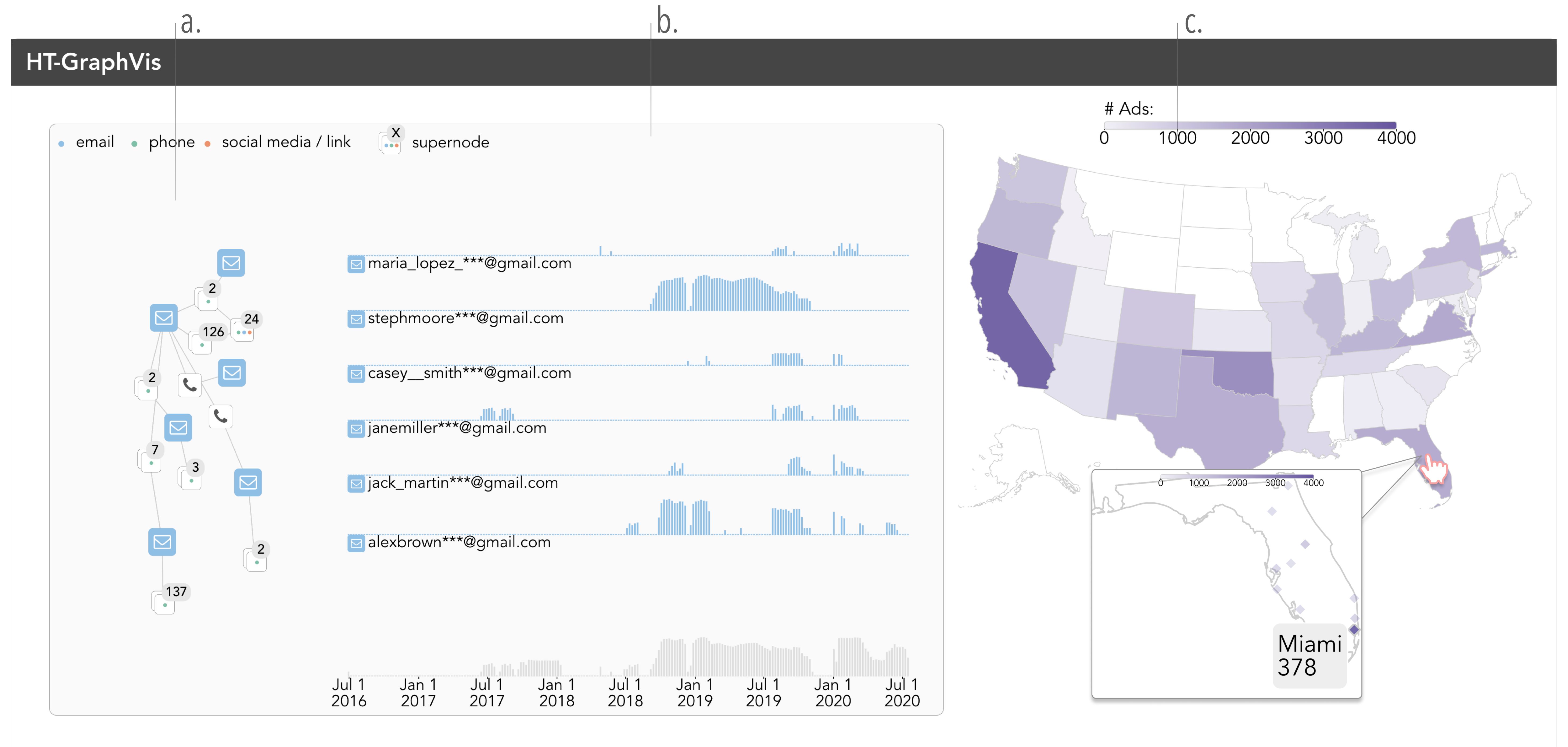


(b) Find central nodes



(c) Find supernodes

2 Fast spatiotemporal exploration. (a) Expert uses *Graph view* to discover connections between metadata and select nodes of interest. (b) Aligned time series in *Timeline view* let experts examine synchronized behavior and select particular weeks. (c) Expert finds hotspots in *Map view* and can **select** a state to zoom in and investigate city-level data.



When an expert **selects** a supernode...

- Central nodes** move to make space for other timeseries
- Expert can scroll through the time series of all contained nodes

